UNIVERSITY OF SURREY ©

Faculty of Engineering and Physical Sciences
School of Mathematics and Physics
Department of Mathematics

Undergraduate Programmes in Mathematics

Module MAT3032; 15 Credits

Advanced Algebra

FHEQ Level 6 (Year 3) Examination

Time allowed: **2 hours** Semester 1 2022/3

Answer **THREE** questions.
If you attempt more than three questions, only your BEST THREE answers will be taken into account.
Show full workings.

Results proved in the module may be assumed and used without proof, unless a proof is requested.

Where appropriate, the mark carried by an individual part of a question is indicated in square brackets [ ], and the maximum mark of this assessment is 75.

Candidates may only use calculators that are non-programmable, with no alphanumeric memory and not wireless enabled.

Additional Material:
one side of A4 paper of handwritten and/or typeset notes, 0 handouts

Question 1

(a) Let H be an abelian subgroup of a finite multiplicative group G with identity element e. For all $g \in G$ and $h \in H$ let $h(g) = hgh$.

(i) Show $(h, g) \mapsto h(g)$ is a group action of H on G. **[7 marks]**

(ii) Show that H_e, the stabiliser of e, is the set of all self inverse elements of H. **[4 marks]**

(iii) Show that the orbit $\mathcal{O}(e)$ is a subgroup of H and that if $|H|$ is odd then $\mathcal{O}(e) = H$. **[6 marks]**

(b) Let σ be a 3-cycle in the symmetric group S_n, where $n \geqslant 3$.

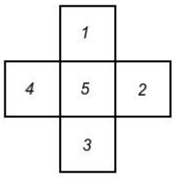
(i) State the cycle type of σ and hence show the order of the centraliser of σ in S_n is $3(n-3)!$. Furthermore, find the order of the conjugacy class of σ in S_n. **[5 marks]**

(ii) Find, with justification, the number of 3 cycles in S_5. **[3 marks]**

[Total for Question 1: 25 marks]

Question 2

(a) A cross is formed from five squares of identical size and side length as shown in the following diagram:



The cross can be rotated in its plane and/or turned over and each square appears the same on both sides.

(i) Write down the symmetries of the figure as permutations of $\{1, 2, 3, 4, 5\}$, where each permutation you write down should be expressed as a product of *disjoint cycles*. **[6 marks]**

(ii) Give a presentation of the symmetry group of the figure and state a group which the symmetry group is isomorphic to. **[5 marks]**

(iii) The square regions are to be coloured using q available colours. Find the number of distinguishable q-colourings of the cross. **[4 marks]**

(b) Let ψ be an automorphism of a multiplicative group G. Suppose H is a normal subgroup of G.

(i) You are given that $\psi(H)$ is a subgroup of G. Show that $\psi(H)$ is normal in G. **[3 marks]**

(ii) Suppose a and b lie in the same (left) coset of H in G. Show that $\psi(a)$ and $\psi(b)$ lie in the same (left) coset of $\psi(H)$ in G. **[2 marks]**

(iii) Suppose $\psi(H) = H$. Show that there is an automorphism $\overline{\psi}$ of G/H given by

$$\overline{\psi}(aH) = \psi(a)H \quad \text{for all } aH \in G/H.$$

[5 marks]

[Total for Question 2: 25 marks]

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Question 3

(a) Let G be a group of order $585 = 3^2 \times 5 \times 13$.

(i) Use the Sylow Theorems to show that G has normal subgroups of orders 5 and 13. Deduce that G is not simple. **[7 marks]**

(ii) Find $\phi(5)$ where ϕ is the Euler-Totient Function and show that a 5-element subgroup of G is contained in the centre of G. **[5 marks]**

(iii) Suppose G is cyclic. Show that G has a subgroup isomorphic to C_9 and conclude G *has no* subgroup isomorphic to $C_3 \times C_3$. **[6 marks]**

(b) Let m and n be positive integers. A group G of order n acts on a set X with m elements. Suppose there is a non-identity element $g \in G$ such that $g(x) = x$ for all $x \in X$ and there is a non-identity element $h \in G$ such that $h(x) \neq x$ for at least one $x \in X$.

(i) Show that $1 < |\tilde{G}_X| < |G|$, where $\tilde{G}_X$ is the stabiliser of X. **[2 marks]**

(ii) Show that n and $m!$ are not coprime, i.e that $gcd(n, m!) > 1$ and that G is not simple. **[5 marks]**

[Total for Question 3: 25 marks]

Question 4

(a) Let u be the element $2+3i+4j+5k$ in the ring $\mathbb{H}$ of real quaternions. Recall that $i^2=j^2=k^2=-1$ and $ij=-ji=k, jk=-kj=i$ and $ki=-ik=j$.

(i) Use u to find integers p,q,r,s such that $(2^2+3^2+4^2+5^2)^2=p^2+q^2+r^2+s^2$. **[6 marks]**

(ii) Find the multiplicative inverse of u, in the form $a+bi+cj+dk$ where $a,b,c,d\in\mathbb{R}$. **[3 marks]**

(b) A ring R has idempotent elements e and f such that $ef=fe$. Show that $e-ef$ is an idempotent element and identify an orthogonal pair of idempotents in R. **[5 marks]**

(c) Let A be the associative algebra with basis $\{1,v,v^2\}$ over the field $\mathbb{F}_2=(\{0,1\},+_2,\times_2)$, such that the element v satisfies the equation $v^3=1+v+v^2$.

(i) State the number of elements in A. **[2 marks]**

(ii) Find the matrix which represents a general element $a+bv+cv^2$ in the regular representation of A relative to $(1,v,v^2)$. **[5 marks]**

(iii) Show that $1+v$ has no multiplicative inverse in A. **[4 marks]**

[Total for Question 4: 25 marks]

EXTERNAL EXAMINER: Ian McIntosh
INTERNAL EXAMINER: James Roberts

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